

Sudipto Sarkar

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Department of Electrical and Electronic Engineering, BUET, Dhaka, Bangladesh

SUMMARY

- Electrical and Electronic Engineering graduate from Bangladesh University of Engineering and Technology (BUET) with research experience in Deep Learning, Computer Vision, Multimodal AI, Speech Recognition, and Foundation Models. Author of peer-reviewed publications and an arXiv preprint, with multiple top placements in IEEE AI competitions. Experienced in PyTorch, TensorFlow, model optimization, multimodal learning, and developing practical AI systems.

EDUCATION

- Bangladesh University of Engineering and Technology (BUET)** 2022 (January) – 2026(Expected)
B.Sc. Engg., Electrical and Electronic Engineering (Level 4) Dhaka, Bangladesh
 - CGPA: 3.21 (up to Level-4 Term 2)
- Notre Dame College** 2018 – 2020
Higher Secondary School Certificate (HSC), Science Dhaka, Bangladesh
- Kushtia Zilla School** 2012 – 2018
Secondary School Certificate (SSC), Science Kushtia, Bangladesh

PROFESSIONAL EXPERIENCE

- Langgol – Agritech** 2025 – present
Founder and Senior Software Developer Dhaka, Bangladesh
- Udvash (Education Institute)** 2023 – Present
Math Instructor Dhaka, Bangladesh
- Somakolon – the learning app** 2020 – 2024
Founder and Senior Developer Dhaka, Bangladesh

PUBLICATIONS

[1] F. Labiba, M. A. Hasan, S. Sarkar, A. Shahriar, N. Tasnim, and S. A. Fattah, “Real-Time Multi-Modal Drone and Bird Tracking Using Modality-Aware Adaptation and Physics-Aware Filtering,” in *2025 IEEE International Women in Engineering (WIE) Conference on Electrical and Computer Engineering (WIECON-ECE)*, Cox’s Bazar, Bangladesh, 2025.

Preprints

[2] S. Sarkar, M. A. Hasan, K. A. Shahriar, F. Labiba, N. Tasnim, and S. A. H. Fattah, “EGD-YOLO: A Lightweight Multimodal Framework for Robust Drone-Bird Discrimination via Ghost-Enhanced YOLOv8n and EMA Attention under Adverse Conditions,” *arXiv preprint arXiv:2510.10765*, 2025.

Under Review

[3] S. Sarkar and H. Imtiaz, “GLFNet: Gated Local-Focus Network for EEG-Based Visual Semantic Decoding,” *Manuscript under review*, 2026.

TECHNICAL SKILLS

- Programming Languages:** Python, C++, C, JavaScript, SQL, PHP, HTML, CSS, MATLAB
- Machine Learning & AI:** PyTorch, TensorFlow, OpenCV, Scikit-learn, Hugging Face Transformers, OpenCLIP, YOLOv8, OCR, Whisper, Wav2Vec2, LoRA, Pandas, NumPy
- Backend & DevOps:** Node.js, Firebase APIs, Docker, Git, GitHub
- Hardware & Robotics:** Arduino, ESP32, Raspberry Pi 4B, Pixhawk, FPGA, ROS
- Tools:** Kaggle, Vast.ai, Visual Studio Code, Blender, Figma, PSS/E, LTspice, Mission Planner, RealVNC, Microsoft Word, Excel

RESEARCH INTERESTS

- Brain-Computer Interface (BCI)
- Computer vision
- Quantum Computing and Quantum Machine Learning
- Deep Learning
- Foundation Models & LLMs
- Biomedical AI

LEADERSHIP AND EXTRA-CURRICULAR ACTIVITIES

• President <i>BUET Robotics Society</i>	2025 – Present
• Vice Chairperson <i>IEEE BUET Student Branch Chapter</i>	2025 – Present
• Senior Coordinator <i>Team Interplanetary - Drone Sub-Team</i>	2025 – Present
• Treasurer <i>BUET Robotics Society</i>	2024 – 2025

RELEVANT COURSEWORK

• AI & ML	• Digital Signal Processing	• Digital Electronics
• Random Signal Processing	• Communication System	• Matlab
• Power Electronics	• Control System	• Robotics and Automation

PROJECTS

• Autonomous Rescue Drone for Locating Survivors  GitHub <i>Computer Vision, AI, UAV</i>	Sep 2024 - Dec 2024
◦ Developed an AI-powered drone for search and rescue in disaster areas.	
◦ Implemented real-time human detection using a tracking algorithm and OpenCV.	
◦ Integrated GPS-based autonomous flight control for structured coverage and waypoint navigation.	
• Deepfake Detection using CLIP-ViT <i>OpenCLIP, ViT, PyTorch</i>	Oct 2024 - Jan 2025
◦ Developed a deepfake detection model using OpenCLIP-ViT trained on a balanced dataset.	
◦ Achieved 98% training and 97% validation accuracy using strict mini-batch training and augmentation.	
◦ Evaluated the model on the IEEE SP Cup 2025 dataset (43k real, 219k fake images).	
• Multi-Modal Political Meme Classification  GitHub <i>Computer Vision, NLP, Multimodal AI, PyTorch</i>	2025
◦ Proposed a novel dual co-attention framework integrating CLIP, XGLM, EasyOCR, and PaddleOCR for multimodal political meme classification.	
◦ Achieved 90.7% validation accuracy and 82.8% test accuracy on a large-scale political meme dataset.	
◦ Enabled interpretable cross-modal feature fusion , improving classification robustness for multilingual political memes.	
• Shabdotori: Bengali Dialect Speech Recognition  GitHub <i>Speech Processing, ASR, Deep Learning, LoRA</i>	2025
◦ Proposed a two-stage Bengali dialect ASR pipeline combining Wav2Vec2-based dialect classification with LoRA fine-tuned Whisper models.	
◦ Achieved significant improvements in WER and CER over dialect-agnostic ASR through region-specific transcription.	
◦ Enhanced recognition robustness using comprehensive audio data augmentation for low-resource Bengali dialects.	
• Brain Tumor Detection on MRI Images  GitHub <i>Digital Image Processing, MATLAB</i>	Dec 2023 - Feb 2024
◦ Developed a CAD system for automated brain tumor detection in MRI images.	
◦ Improved MRI image clarity through advanced image processing and filtering techniques.	
◦ Achieved high sensitivity and specificity in tumor detection (results from validation experiments).	
• FPGA Based Home Automation System  GitHub <i>Quartus, FPGA, Verilog</i>	Sep 2024 - Jan 2025
◦ Implemented an IoT-based home automation system on FPGA using Quartus and Verilog.	
◦ Device regulation in a room controlled by the number of individuals (sensor-driven logic).	
• Underfrequency Load Shedding Implementation  GitHub <i>PSS/E</i>	Dec 2023 - Mar 2024
◦ Analyzed underfrequency load shedding schemes using PSS/E on the IEEE 24-bus test system.	
◦ Simulated load shedding strategies and assessed system recovery and stability.	

- **Ultrasonic Obstacle Mapping Bot** |  [GitHub](#) 2025
Embedded Systems, Robotics, IoT
 - Developed an autonomous robot using NodeMCU and HC-SR04 ultrasonic sensor for real-time indoor mapping.
 - Implemented obstacle avoidance and 180° environment scanning using a servo-based sensing mechanism.
 - Designed a web-based interface to visualize mapped surroundings dynamically.
 - Built as a low-cost, educational DIY platform for robotics and embedded systems experimentation.

THESIS

- **NeuroVision: EEG-to-Video Generation using Multimodal Deep Learning** 2025–present
Brain signal driven video generation and cross-modal representation learning

HONORS AND AWARDS

- **2nd Runner Up: IEEE VIP-CUP 2025** 2025
Designed a lightweight multimodal drone–bird detection and tracking system for the IEEE VIP-CUP international competition
 - **1st Runner Up: IAS Humanatarian Contest 2025** 2025
Autonomous Disaster response drone for search and rescue and medical supply delivery
 - **1st Runner Up: Project Showcasing** 2024
EEE DAY 2023
 - **2nd Runner Up: Project Showcasing** 2025
Intra BUET Robo Challenge
 - **1st Runner Up: IEEE-WIE Robotics for Climate Change** 2024
WIE BD Summit
 - **Champion: University Innovation Hub Programme 2025** 2025
IC4, BUET
 - **Finalist: Orange Corners Bangladesh Ideation Challenge** 2025
IC 5.0
 - **Finalist: Datathon CUET CSE FEST 2025** 2025
Developed a novel pipeline using paddle-OCR to classify meme as political and non-political
 - **Finalist: Hult Prize at BUET** 2024

ADDITIONAL INFORMATION

Languages: Bengali (Native), English (Fluent), Hindi

Interests: Robotics, Quantum Physics, Technology Innovation, Data Security, Education Science Communication